

Guarding Self-Evolving AI with Proofs

llcore research findings 2026 — verified neural evolution on CPU

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github.com/furuse-kazufumi/llcore (paper draft + all experiment code/data public)

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Problem: on what grounds may a self-evolving AI admit its own changes?

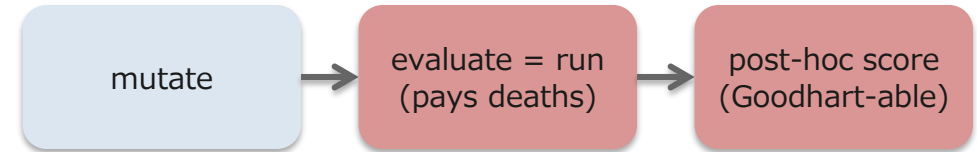
AI now evolves its own core (memory, dynamics)

Mainstream (DGM / SEAL) = empirical gates:
post-hoc scores & past failures

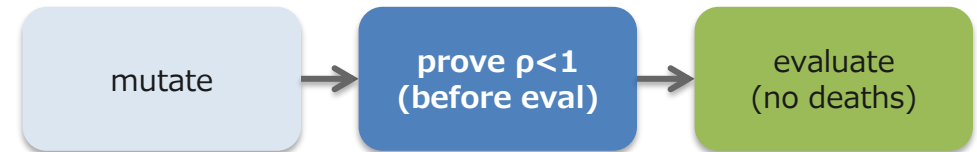
Structural weakness: (1) learns by dying (2)
Goodhart-able

llcore: narrow the property, and a proof can gate
before evaluation

Empirical gate (DGM/SEAL line)



Proof gate (llcore)



Unprovable changes rejected before evaluation (fail-closed)

Approach: a sound contraction-certifier gate (fail-closed)

Certified property = contraction ($\rho < 1$): internal state cannot diverge

Proofs are theorem-derived = non-fabricable (unlike scores)

Gated in-loop, before evaluation — 17-19x cheaper than post-hoc

All experiments pre-registered; negatives disclosed at full strength

Three sound certifier rungs (stronger downward)

closed-form ∞ -norm — $O(n^2)$, no solver

vertex 2-norm (SVD)

Lyapunov SDP (CLARABEL)

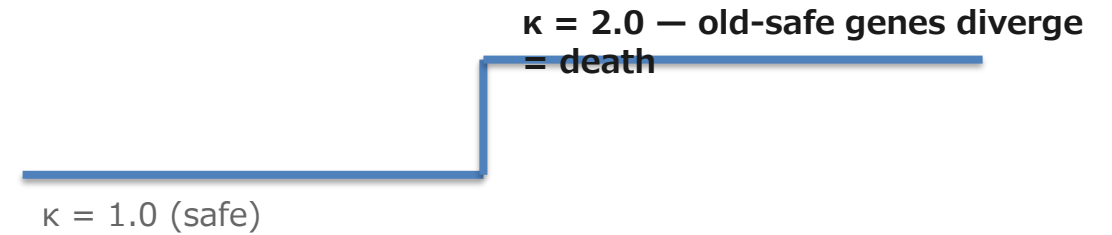
no proof → rejected (fail-closed)

Testbed: a lethal environment x three roads from experience to memory

The environment steps κ by 2x $\rightarrow$ previously-safe genes diverge = death

Death is a real, unavoidable threat (cannot be routed around)

Measured in the steady state after warm-up (n=20 seeds, pre-registered)



ENDO
foresight
(proof)

REVIVE
die $\rightarrow$ repair $\rightarrow$ revive

OBSERVE
learn from
others' deaths

+ baselines: NONE (no gate) / EXO (design-time frozen gate)

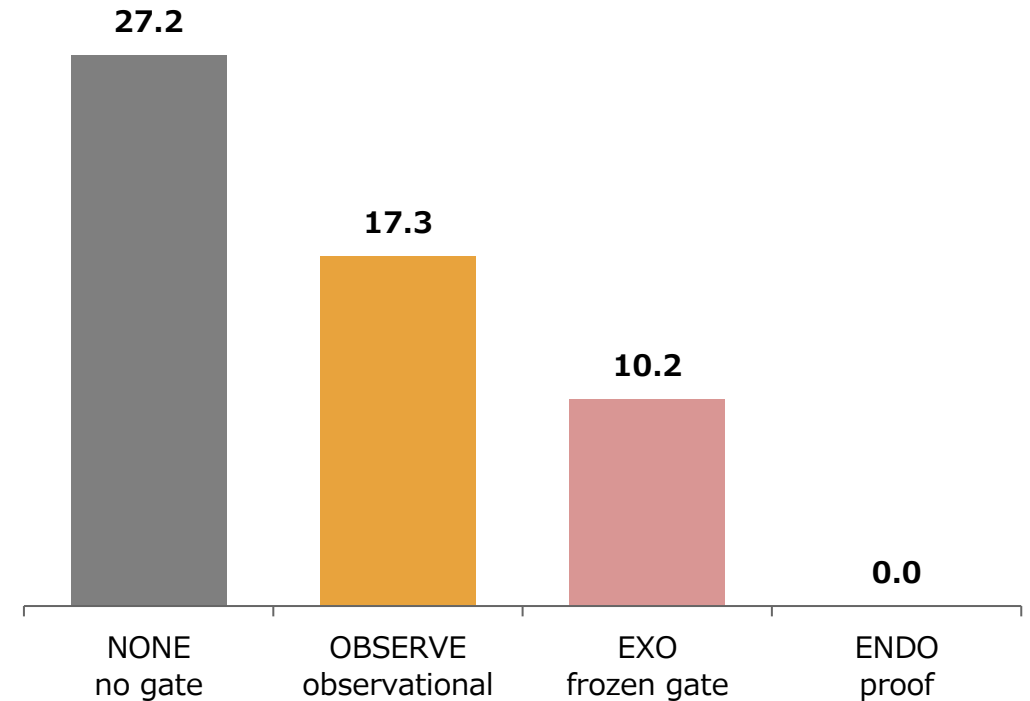
Finding 1: proofs know before dying — zero steady-state deaths

The proof (ENDO) eliminates lethal evaluations entirely

Observational learning works — but deaths remain (17.3). Learning-by-dying is structural

Robust in 12/12 pre-registered configurations (seed/ κ /dimension/difficulty)

Memory capability is not sacrificed (worst fitness gap 0.7%)



Lethal evaluations (steady state, linear substrate, n=20 mean) — data: results_viability_ab.json

$p < 0.001$ (paired sign-flip, n=20)

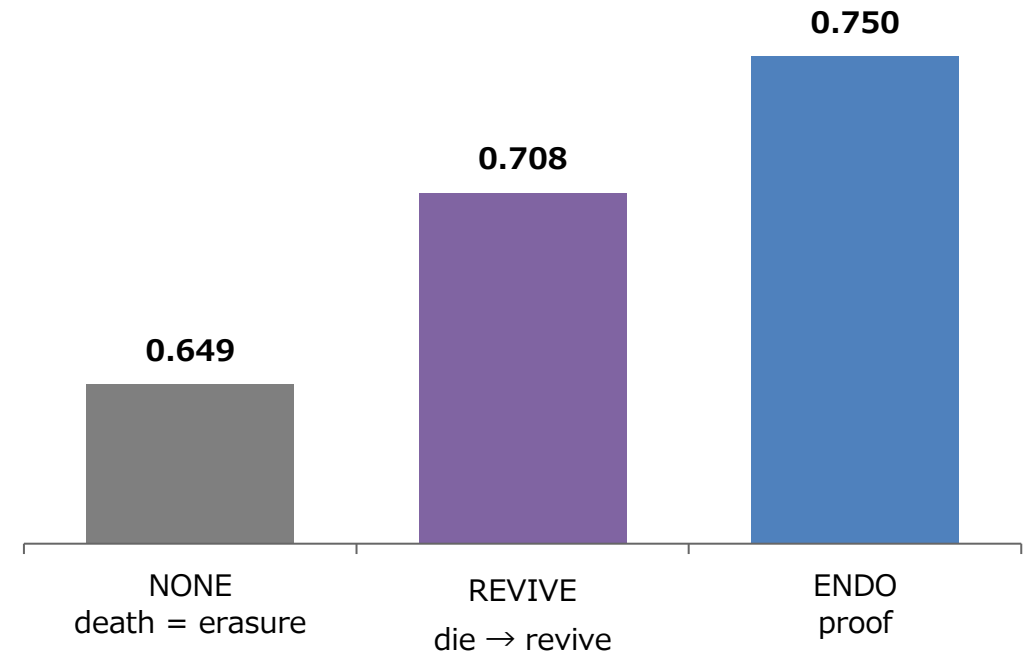
Finding 2: without revival, experience does not become memory

Death removes the individual — and erases its accumulated experience

REVIVE dies about as often as no-gate, yet preserves more population memory

$\Delta = +0.060$ ($p = 0.0011$) — repair preserves the memory channel while safing the dynamics

Death becomes a scar, not a void



Population-mean fitness (linear). Y-axis starts at 0.60 (to make the gap visible; mind the effect size)

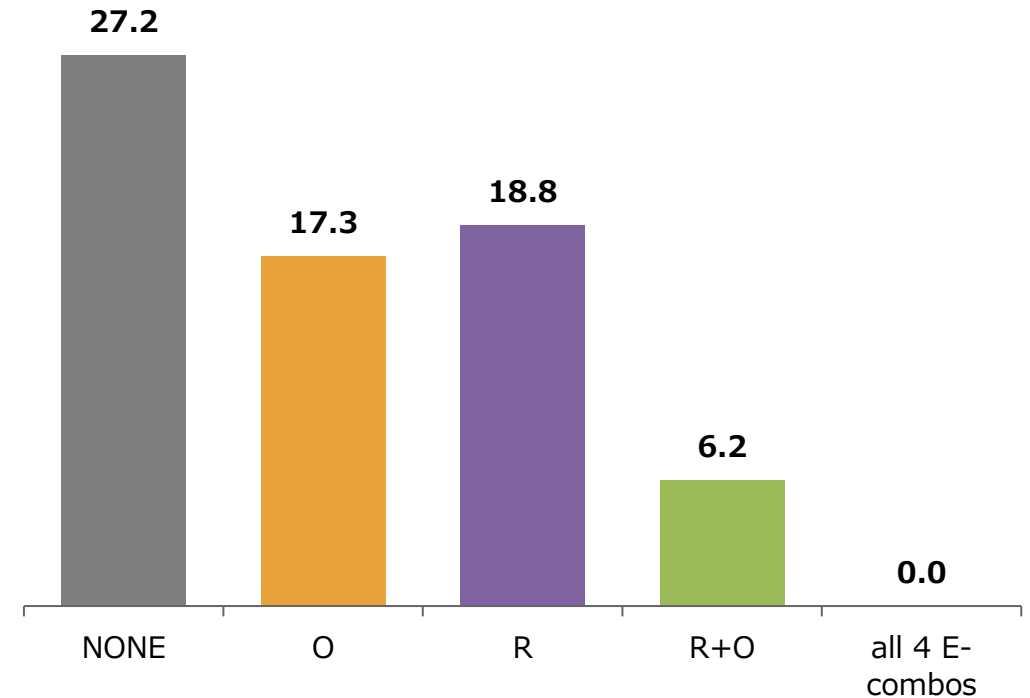
Finding 3: one proof beats stacked empirical mechanisms

All 8 combinations compared under identical conditions (2^3 factorial)

All 4 proof-containing combos: zero deaths

Best proof-free combo = revival+observation (6.2)
— better than alone, never zero

With the proof, empirical mechanisms become redundant (interaction $p=0.0003/0.0045$)



Lethal evaluations (2^3 factorial, linear, $n=20$ mean). E=proof / R=revival / O=observation

Finding 4 (limits): a proof is only sound while its premises hold

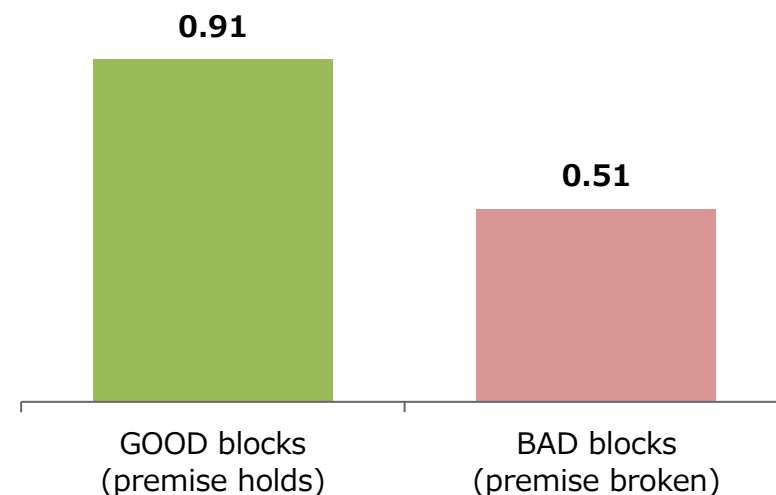
Break the verifier's sensing, and the certificate admits a lethal band

A trust controller on observable evidence does track the regime (figure)

But hedging pays only where violations are costly — a failed hypothesis is reported as-is

Open question: who certifies the certifier's premises?

Verifier under-senses κ by 40% (premise broken)
→ a regime where the proof lies



META's trust in the proof (mean, linear). Tracks the regime from admitted-then-died evidence alone

PoC scale (scalar genes, n=20) — not yet shown at real-LLM scale

Map: how this differs from the nearest work (primary sources verified)

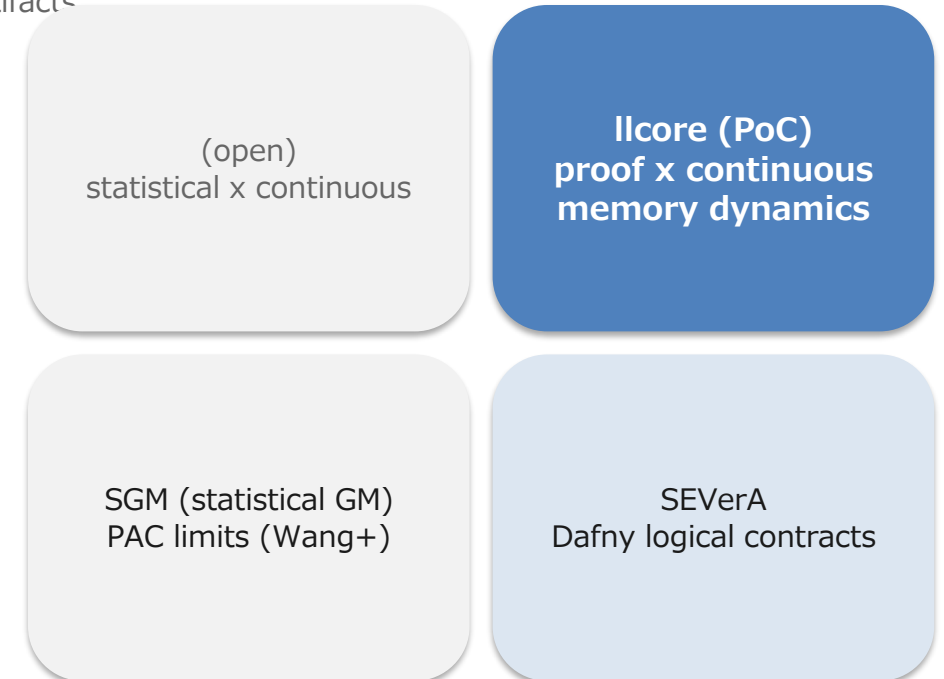
SGM: statistical certificate over discrete task scores — probabilistic

SEVerA: sound Dafny logical contracts — discrete I/O artifacts

PAC limits (Wang+): generalization bounds — not stability

Ilcore occupies the top-right quadrant (proof x continuous memory dynamics) at PoC scale

object: ↑ continuous internal dynamics / ↓ discrete tasks & artifacts



guarantee: ← statistical/empirical | sound deductive proof →

All three verified against primary sources (with adversarial review)

Using this work

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Code / paper draft: Apache-2.0 + Commercial dual license

Closed-source integration / SLA / NDA consulting:
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